KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2022
Department of Computer Science and Engineering
CSE 4115
Computer and Network Security

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Suppose to withdraw money from your Bank account when you put a signature over your Bank cheque, what are the properties essential to retain concerning it? Explain them. Now, analyze that RSA algorithm is capable to fulfill those properties or not. Establish your intuition relating the specific portion of RSA algorithm. (13)
   b) Differentiate between 'Cryptography' and 'Cryptanalysis'. Justify them, i.e., how they are associated with Caesar cipher technique. (10)
   c) List the trends in case of any good cipher. Analyze relevancy of 'Known-Plaintext-Attack' with RSA encryption algorithm. (12)

2. a) Discuss major goals (i.e., CIA) of security. Using a diagram state and explain ingredients of a cryptosystem. (09)
   b) Explain properties of cryptographic hash function. How and why data encryption and hash techniques need to perform concurrently? Propose an outcome from your thinking. (11)
   c) Discuss pros and cons of 'Stream cipher' and 'block cipher'. Assess the working procedures of Caesar cipher, RSA cipher, Vernam cipher etc., with block and Stream ciphers. (15)

3. a) Relating order, prime and primitive element, demonstrate a bonding. Also, judge your bonding for a prime 13, and order 2. (08)
   b) Explain how triple DES performs data encryption. Using a block diagram illustrate the major steps of each cycle of AES algorithm. (11)
   c) Discuss characteristics maintained by "Good Viruses". (06)
   d) Considering two messages m_1 and m_2, develop a model of ElGamal product cipher. Also demonstrate the decryption technique. Step wise write the whole procedure. (10)

4. a) Design a procedure to develop data encryption and digital signature for RSA algorithm together. (08)
   b) "ElGamal digital signature is based on discrete logarithm"—how discrete logarithm manages key generation, signing, verification, etc.? Explain. (11)
   c) How ElGamal cipher can be modified to re-randomize itself? Explain. (07)
   d) Discuss various aspects that make computer networks vulnerable. (09)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Explain relocation of address with example. What are the complementary goals to protect object of OS? (09)
   b) Discuss how ECC works based on the basic of trapdoor function. (10)
   c) Explain the types of Salami attack. What are the impacts and preventions of DHCP spoofing attack? (10)
   d) Differentiate among black hat, white hat and grey hat hackers. (06)

6. a) What is polymorphic virus? How does an Email impersonation attack work? (08)
   b) What are the considerations before committing to a hybrid firewall solution? Let, a businessman works with a public website, accessible from the internet, which put the entire internal network at high risk. Explain which network can help him to conduct his business securely and how? (15)
   c) Explain with figure how paged segmentation removes the problems associated with segmentation and paging. (12)

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7. a) Discuss the problems created by format string attack and countermeasures for them. (15)
   b) Demonstrate the ways to keep devices safe from wiretapping. (12)
   c) Which one is preferable, RSA or, ECC? Explain why? (08)

8. a) Suppose an attacker may introduce extra code, sending new instructions to the application to gain access to IT systems. If attackers know the memory layout of a program, they can intentionally overwrite the executable code with their own code. What kind of attack is this? What are the countermeasures for it? (15)
   b) Explain how documents virus works in Microsoft's excel Microsoft's word file using DDE protocol. (10)
   c) What are the responsibilities of firewall administrators? What are the challenges of intrusion detection system? (10)



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